

Biotin (Vitamin B₇)

Dosage & Timing Guide

Why deficiency is rare, what the hair/skin/nails evidence actually shows, and the FDA safety warning every high-dose user should know.

Evidence-based approach: Institute of Medicine (1998) DRI report; Patel et al. (2017), Skin Appendage Disorders; FDA Safety Communication (2019).

Why Deficiency Is Rare

Biotin is a water-soluble B vitamin involved in fatty acid synthesis and amino acid metabolism. True deficiency is uncommon because biotin is present in a wide range of foods and the body recycles it efficiently. Documented cases center on a small set of causes: eating large amounts of raw egg white over time (avidin, a protein in raw whites, binds biotin and blocks absorption), certain anticonvulsant medications, prolonged IV nutrition without biotin added, and the rare genetic condition biotinidase deficiency, which is screened for at birth in most countries.

Dosing At A Glance

Goal	Dose	Notes
General adequacy (RDA)	30mcg/day	Easily met through diet alone
Pregnancy	30mcg/day	35mcg/day while breastfeeding
Hair/skin/nails (marketed)	2,500-10,000mcg/day	Evidence strongest only if actually deficient
Documented deficiency	Per physician guidance	Often paired with underlying cause treatment

The FDA Safety Warning

In November 2019, the FDA issued a safety communication warning that biotin, even at supplemental doses, can interfere with certain lab tests by causing falsely high or falsely low results. The most serious case involves troponin tests used to diagnose heart attacks in the emergency room — biotin can suppress a troponin reading enough to mask a true cardiac event. It can also distort thyroid panels and some hormone assays. Always tell any doctor or lab technician if you take biotin, and ask how long before bloodwork you should stop, since the window depends on your dose and the specific test.

Quality Checklist

- ☐ Single-ingredient product, not buried in a proprietary hair/skin/nails blend
- ☐ Third-party tested (USP, NSF, or Informed Choice)
- ☐ No megadose without a documented reason to take one
- ☐ Disclosed to your doctor before any scheduled bloodwork

Sources

Institute of Medicine (1998). Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline. National Academies Press.

Patel, D.P. et al. (2017). A Review of the Use of Biotin for Hair Loss. Skin Appendage Disorders, 3(3), 166-169.

U.S. Food and Drug Administration (2019). The FDA warns that biotin may interfere with lab tests: FDA Safety Communication.